

$$\text{Let, } A^{in} = \begin{bmatrix} 0 & 1 & \dots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \quad \text{and Cell number, } N = \begin{bmatrix} n_1 & n_1 & \dots \\ n_2 & n_2 & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}_{N_{CT} \times N_{mets}}$$

$$\text{Consumption matrix, } C = A^{in} \odot N = \begin{bmatrix} 0 & n'_{1,2} & \dots \\ \vdots & \ddots & \dots \\ n'_{N_{CT},1} & n'_{N_{CT},2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}, \text{ where } n'_{i,j} = A^{in}_{i,j} n_i$$

$$\text{Let, } \tau = \text{Column sum}(C) = [T_1 \quad T_2 \quad \dots]_{1 \times N_{mets}}, \text{ where } T_j = \sum_{i=1}^{N_{CT}} n'_{i,j}$$

$$\text{and } \tau' = \begin{bmatrix} \frac{1}{T_1} & \frac{1}{T_2} & \dots \end{bmatrix}_{1 \times N_{mets}}$$

Normalised consumption matrix, $C_{norm} = C \odot \tau'$

$$= \begin{bmatrix} 0 & n'_{1,2} & \dots \\ \vdots & \ddots & \dots \\ n'_{N_{CT},1} & n'_{N_{CT},2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \odot \begin{bmatrix} \frac{1}{T_1} & \frac{1}{T_2} & \dots \end{bmatrix}_{1 \times N_{mets}} = \begin{bmatrix} 0 & \frac{n'_{1,2}}{T_2} & \dots \\ \vdots & \ddots & \dots \\ \frac{n'_{N_{CT},1}}{T_1} & \frac{n'_{N_{CT},2}}{T_2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}}$$

Let Diet, $D = \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$

Net Intake, $I = C_{norm} D$

$$= \begin{bmatrix} 0 & \frac{n'_{1,2}}{T_2} & \dots \\ \vdots & \ddots & \dots \\ \frac{n'_{N_{CT},1}}{T_1} & \frac{n'_{N_{CT},2}}{T_2} & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$$

$$= \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}, \text{ where } \sigma_i = \sum_{j=1}^{N_{mets}} D_j \frac{n'_{i,j}}{T_j}$$

$$A^{out} = \begin{bmatrix} 0 & 1 & \cdots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \cdots \end{bmatrix}_{N_{CT} \times N_{mets}}$$

Inverse out-degree, $\Omega = \begin{bmatrix} \omega_1^{-1} \\ \omega_2^{-1} \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$

Byproduct fraction, $f = 0.5$

Output, $S = A^{out} \odot (f \ \Omega)$

$$= \begin{bmatrix} 0 & \frac{f}{\omega'_{1,2}} & \cdots \\ \vdots & \ddots & \vdots \\ \frac{f}{\omega'_{N_{CT},1}} & \frac{f}{\omega'_{N_{CT},2}} & \cdots \end{bmatrix}_{N_{CT} \times N_{mets}}, \text{ where } \frac{1}{\omega'_{i,j}} = \frac{1}{\omega_i} A^{out}_{i,j}$$

Secreted metabolome part, $M^s = S^T I$

$$= \begin{bmatrix} 0 & \dots & \frac{f}{\omega'_{N_{CT}, 1}} \\ \frac{f}{\omega'_{1,2}} & \dots & \frac{f}{\omega'_{N_{CT},2}} \\ \vdots & \dots & \vdots \end{bmatrix}_{N_{mets} \times N_{CT}} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$$

$$= \begin{bmatrix} \sum_{i=1}^{N_{CT}} \sigma_i \frac{f}{\omega'_{i, 1}} \\ \sum_{i=1}^{N_{CT}} \sigma_i \frac{f}{\omega'_{i, 2}} \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$$

Net predicted metabolome, $M^{pred} = M^{unused} + M^s$,

where $\forall k \in \{1, 2, \dots, N_{mets}\}$, $M_k^{unused} = D_k \forall \sum_{i=1}^{N_{CT}} \frac{f}{\omega'_{i,k}} = 0$

and 0 otherwise

Toy example

Let, $A^{in} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}_{2 \times 3}$ and Cell number, $N = \begin{bmatrix} 100 & 100 & 100 \\ 200 & 200 & 200 \end{bmatrix}_{2 \times 3}$

Consumption matrix, $C = A^{in} \odot N = \begin{bmatrix} 0 & 100 & 100 \\ 200 & 0 & 200 \end{bmatrix}_{2 \times 3}$

Let, $\tau = \text{Column sum}(C) = [200 \quad 100 \quad 300]_{1 \times 3}$ and $\tau' = \begin{bmatrix} \frac{1}{200} & \frac{1}{100} & \frac{1}{300} \end{bmatrix}_{1 \times 3}$

Normalised consumption matrix, $C_{norm} = C \odot \tau' = \begin{bmatrix} 0 & 100 & 100 \\ 200 & 0 & 200 \end{bmatrix}_{2 \times 3} \odot \begin{bmatrix} \frac{1}{200} & \frac{1}{100} & \frac{1}{300} \end{bmatrix}_{1 \times 3} = \begin{bmatrix} 0 & 1 & \frac{1}{3} \\ 1 & 0 & \frac{2}{3} \end{bmatrix}_{2 \times 3}$

Let Diet, $D = \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}_{3 \times 1}$

Net Intake, $I = C_{norm} D = \begin{bmatrix} 0 & 1 & \frac{1}{3} \\ 1 & 0 & \frac{2}{3} \end{bmatrix}_{2 \times 3} \begin{bmatrix} 10 \\ 20 \\ 25 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 28.33 \\ 18.33 \end{bmatrix}_{2 \times 1}$

$$A^{out} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}_{2 \times 3}$$

$$\text{Inverse out-degree, } \Omega = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{1} \end{bmatrix}_{2 \times 1}$$

$$\text{Byproduct fraction, } f = 0.5$$

$$\text{Output, } S = A^{out} \odot (f \Omega)$$

$$= \begin{bmatrix} \frac{0.5}{2} & \frac{0.5}{2} & 0 \\ \frac{0.5}{1} & 0 & 0 \end{bmatrix}_{2 \times 3} = \begin{bmatrix} 0.25 & 0.25 & 0 \\ 0.5 & 0 & 0 \end{bmatrix}_{2 \times 3}$$

$$\text{Secreted metabolome part, } M^s = S^T I = \begin{bmatrix} 0.25 & 0.5 \\ 0.25 & 0 \\ 0 & 0 \end{bmatrix}_{3 \times 2} \begin{bmatrix} 28.33 \\ 18.33 \end{bmatrix}_{2 \times 1}$$

$$= \begin{bmatrix} (0.25 \times 28.33) + (0.5 \times 18.33) \\ (0.25 \times 28.33) + (0 \times 18.33) \\ 0 + 0 \end{bmatrix}_{3 \times 1} = \begin{bmatrix} 16.2475 \\ 7.0825 \\ 0 \end{bmatrix}_{3 \times 1}$$

Net predicted metabolome, $M^{pred} = M^{unused} + M^s$,

$$= \begin{bmatrix} 0 \\ 0 \\ 25 \end{bmatrix}_{N_{mets} \times 1} + \begin{bmatrix} 16.2475 \\ 7.0825 \\ 0 \end{bmatrix}_{N_{mets} \times 1}$$

$$= \begin{bmatrix} 16.2475 \\ 7.0825 \\ 25 \end{bmatrix}_{N_{mets} \times 1}$$